

UKAS QUALITY MANAGEMENT 005

SYSTEM CERTIFICATION ISO 9001:2008 SGS

ACROW

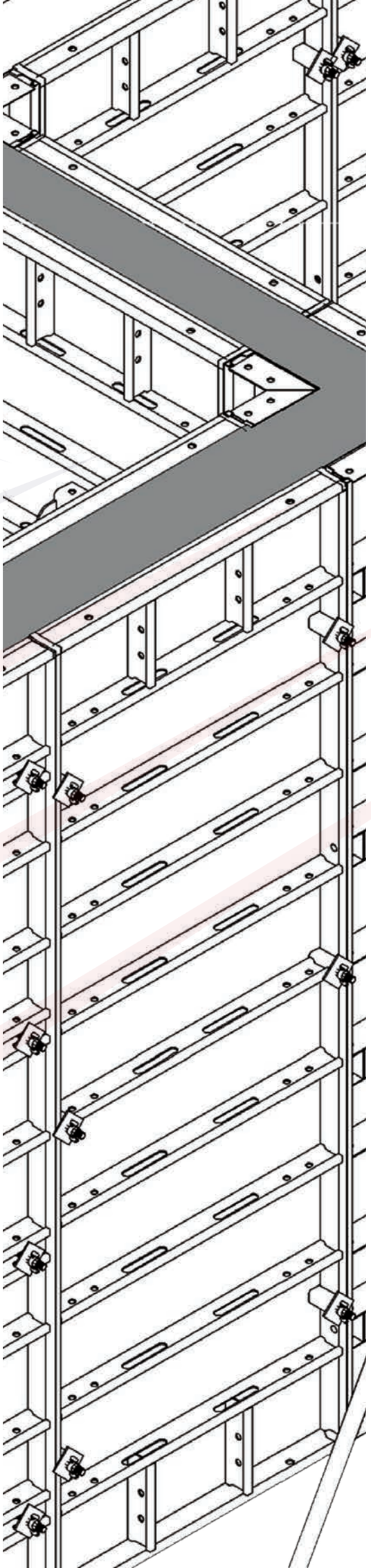
Formwork Technology

ECO- Form System

ACROW MISR Provides Full Site Service Based On More Than 35 Years Experience In The Field

ACROW

Formwork Technology



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ACROW
Formwork Technology

Product Description

Being one of the largest formwork and scaffolding companies in Egypt, We are introducing the ECO form light weight panel system. It is an innovative system where a range of panels and accessories are responsible for quick and economical framing of any vertical concrete shape in the buildings. Moreover, Due to the lightness and high materials quality of the ECO form system, it allows for either manual or crane handling which enables for extra-ordinary performance.
In order to reach the optimum balance between both the lightness and the strength, new materials and technical innovations were introduced and combined together resulting for this ideal balance. The new materials used in this system were as follows:

1. High engineered light weight steel profile which provides a more durable and economical system.
2. Plywood covered by phenolic film and standard accessories such as clamps in order to make gangs by joining panels.

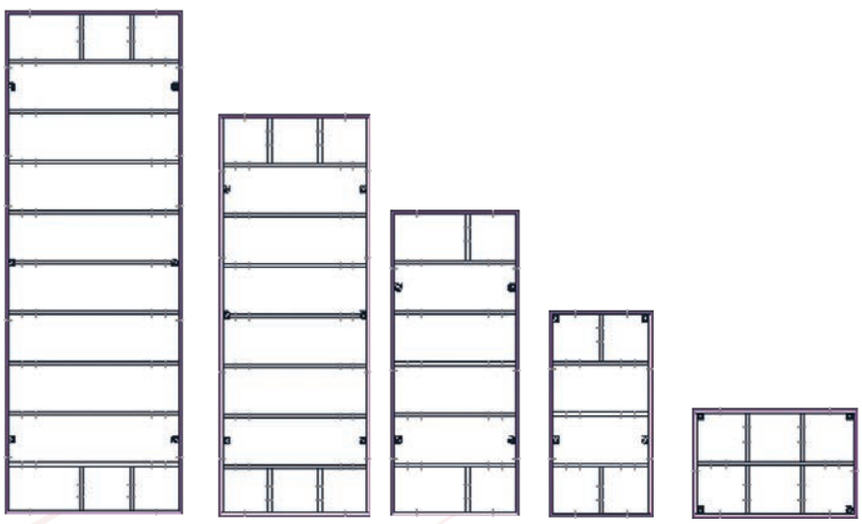
Eco form is the best solution for foundations, cellar tanks, low retaining walls, parapets and small areas. This perfection returns mainly to the sturdy elements which are actually light in weight and handy to enable Eco form to be on the top solutions to be used on these applications. Moreover, what makes it easier for being used in high walls is the flexibility of the panels to be stacked upright or sideways to fit the intended purpose smoothly.





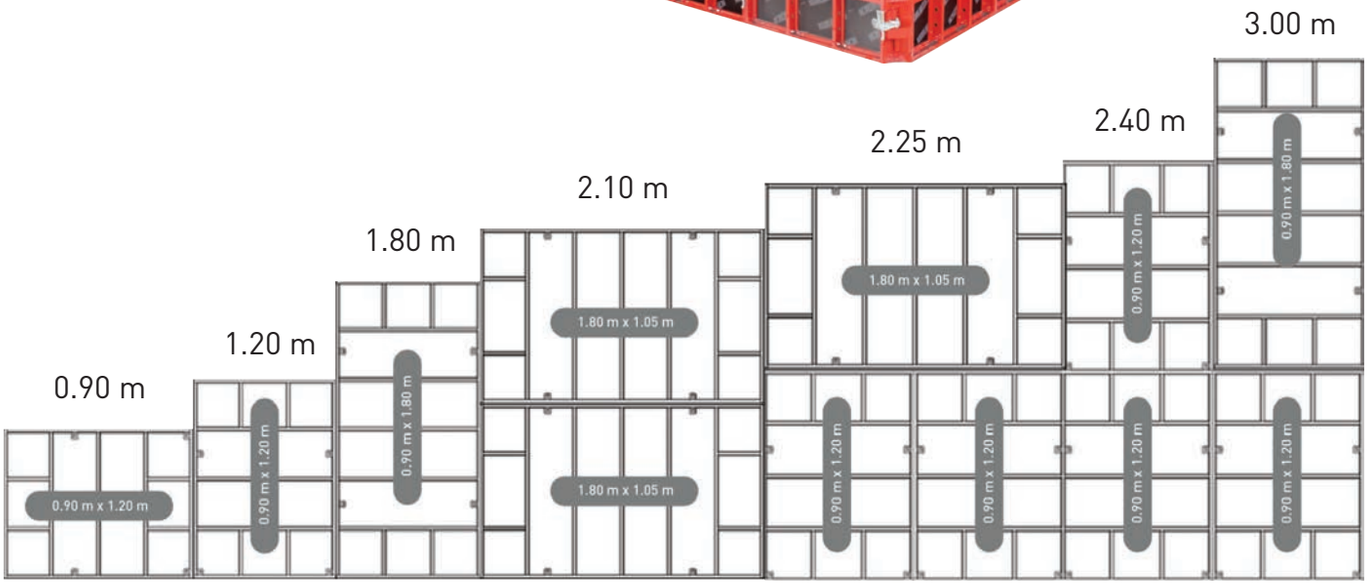
Additional Features

- Variable panel sizes to suit different structures with different shapes (Walls, Columns, & foundations).
- Self-aligning accessories reducing both the erection times and cost.
- High quality 15 mm finishing plywood enabling the user to use it for more than 70 times as sheeting.
- Superb quality of powder coating finish providing the system with a unique protection level which actually results in longer lifespan.
- Permissible concrete pressure is 60 KN/m² for columns and 45 KN/m² for walls.



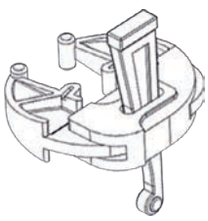


Height Adjustment



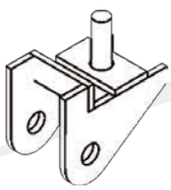
Eco-Form Formwork Accessories

ECO-Form Clamp (EFC)



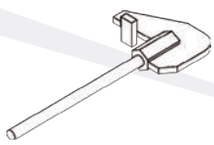
- EFC primarily creates fast, self-aligning and tension proof joints between panels. EFC doesn't require the use of any other internal panel connections; it is simply the non-loose parts EFC and a hammer to perform the job.

ECO-Form Shore Connector



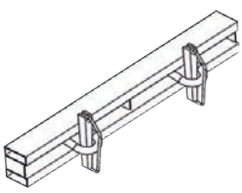
- Eco-form shore connector connects the push pull props to the Eco-form panels. In addition, it uses hex nut fixed to the horizontal ribs of the panels providing the system with much more flexibility.

ECO-Form Stop End Tie



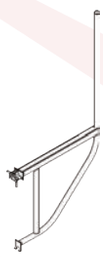
- Eco-form stop end tie connects the compensation waler to the eco-form panel which supports the forming timber that is used as a stop end.

Compensation Waler



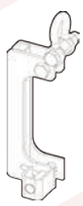
- Compensation Waler fills in closures up to 300 mm in which it transfers the tie forces and aligns the panels at the same time. Special thanks to the movable hooking system which enables forming any closure up to this pre-defined range.

Eco-Form access bracket



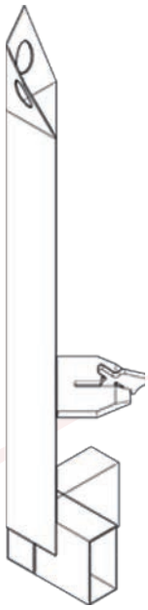
- Since safety during concreting is a must, the wide working access bracket enables for a safe and faster working environment with an allowable load of 150 KG/m².

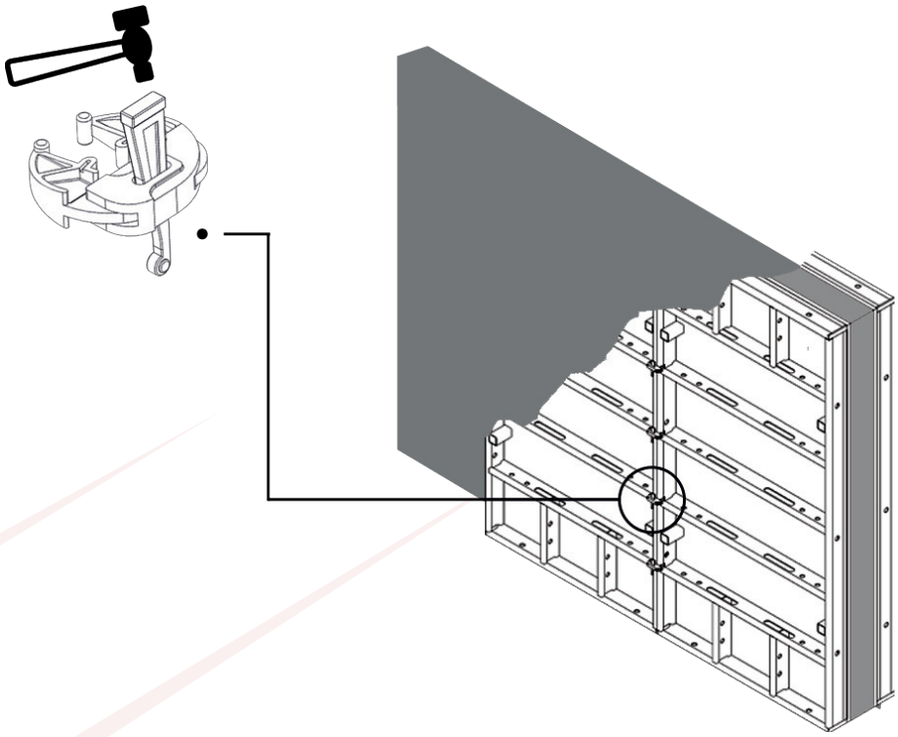
Eco-form lifting hook

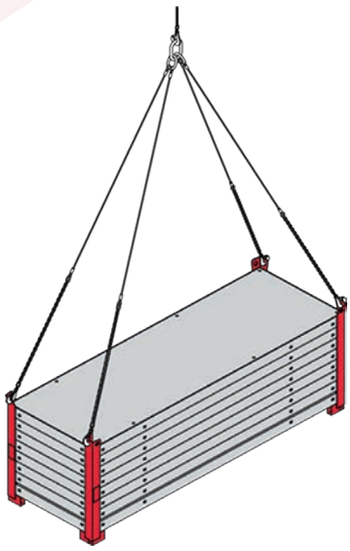


- Eco form lifting hook enables the use of 2 units either individual panels or large gang forms (up to 20 m²) where it also ensure a safely lifting and moving (by crane) which actually results in a dramatically saving in both the time and effort. Special thanks to the self-locking system where it enable automatic hook locking after being attached to the panel. The allowable load for the eco-form lifting hook is 750 KG/ Hook and maximum of 60º between lifting chains.

ECO - Form Formwork Accessories



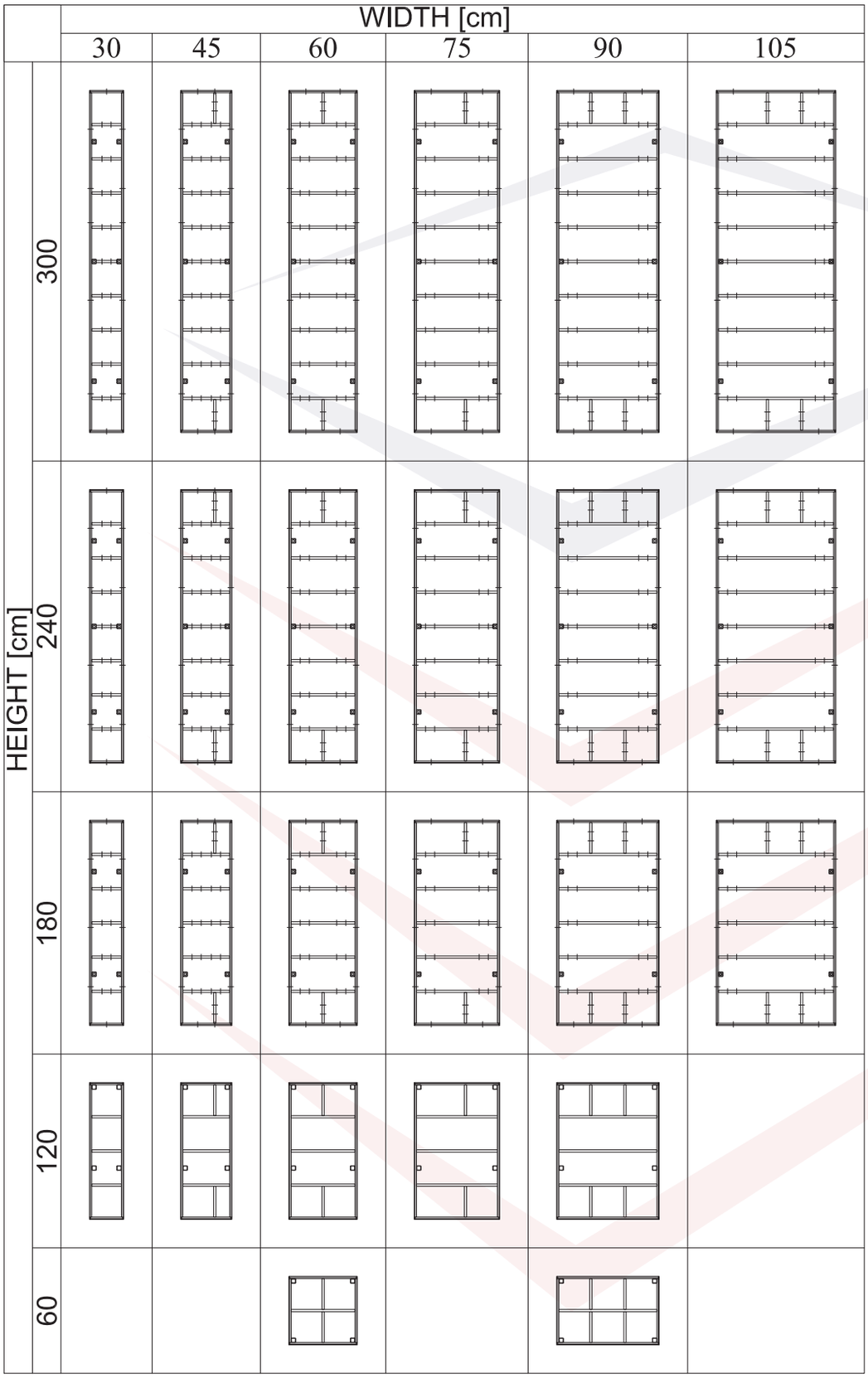




Stacking Device

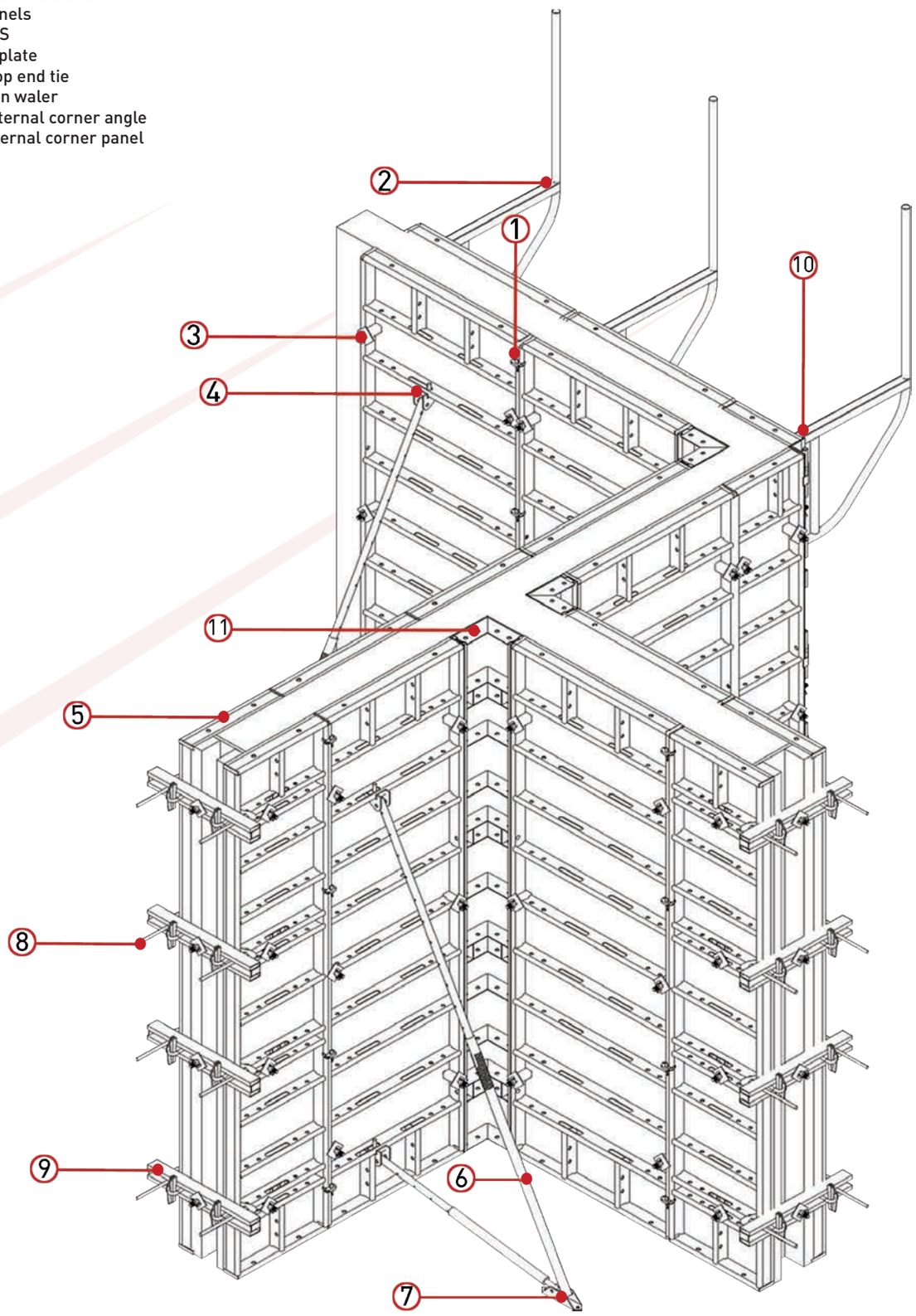
- Stacking device ease the crane handling and ensures easier site stacking. Moreover, the stacks can be easily moved with a fork-lift. The stacking device enables quick and safe sticking of 8 panels of a particular size.

ECO - Form Wall Panels



Wall Overview

1 Eco-Form Coupler (EFC)
2 Access Bracket
3 DW 16 tie system
4 Eco-Form Shore connector
5 Eco-Form panels
6 Push Pull PPS
7 Double base plate
8 Eco-Form stop end tie
9 Compensation waler
10 Eco-Form external corner angle
11 Eco-Form internal corner panel

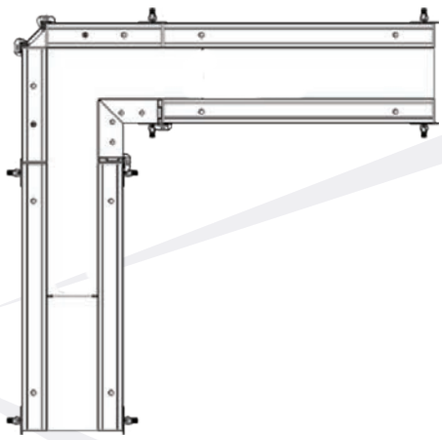


Corners

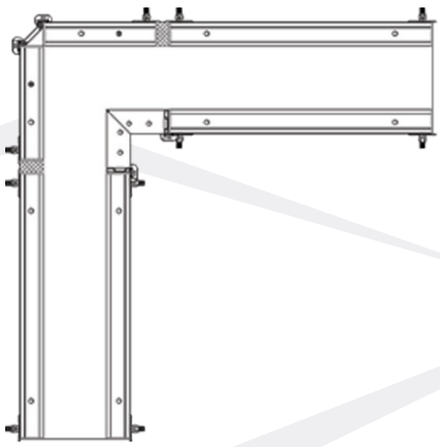
Eco-form panels and accessories ease the formation of corners with different wall thicknesses.



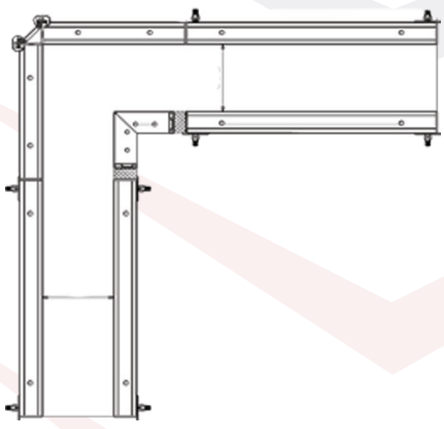
Wall Thickness 200 mm



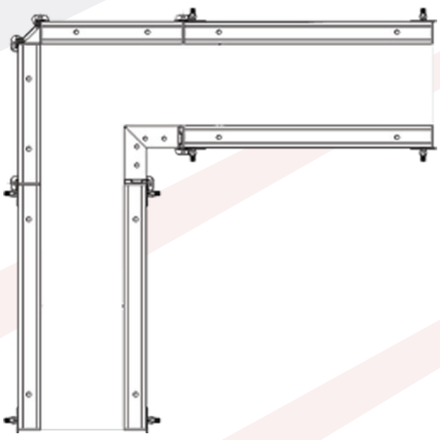
Wall Thickness 250 mm

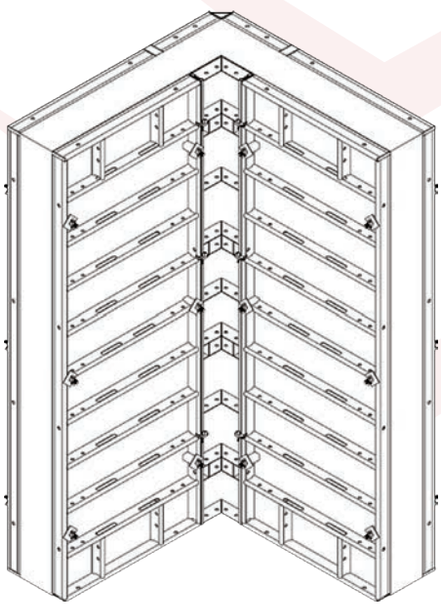


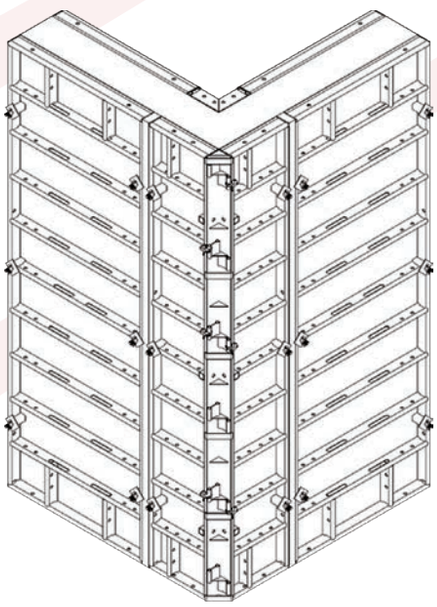
Wall Thickness 300 mm



Wall Thickness 350 mm



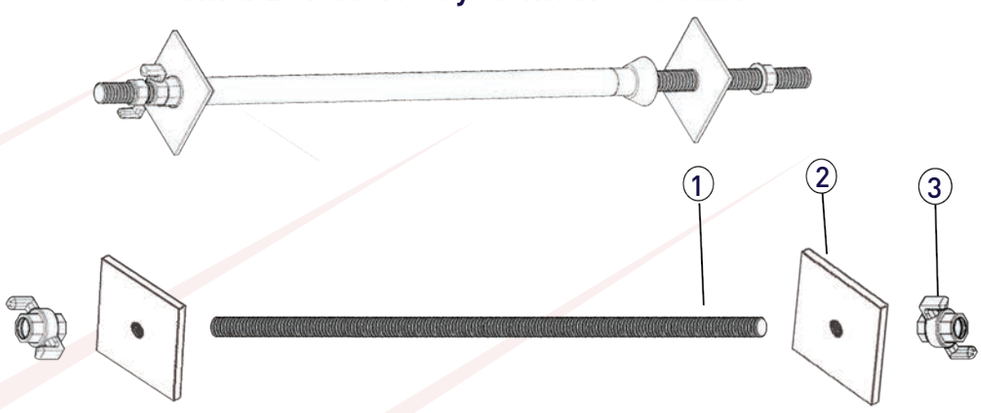




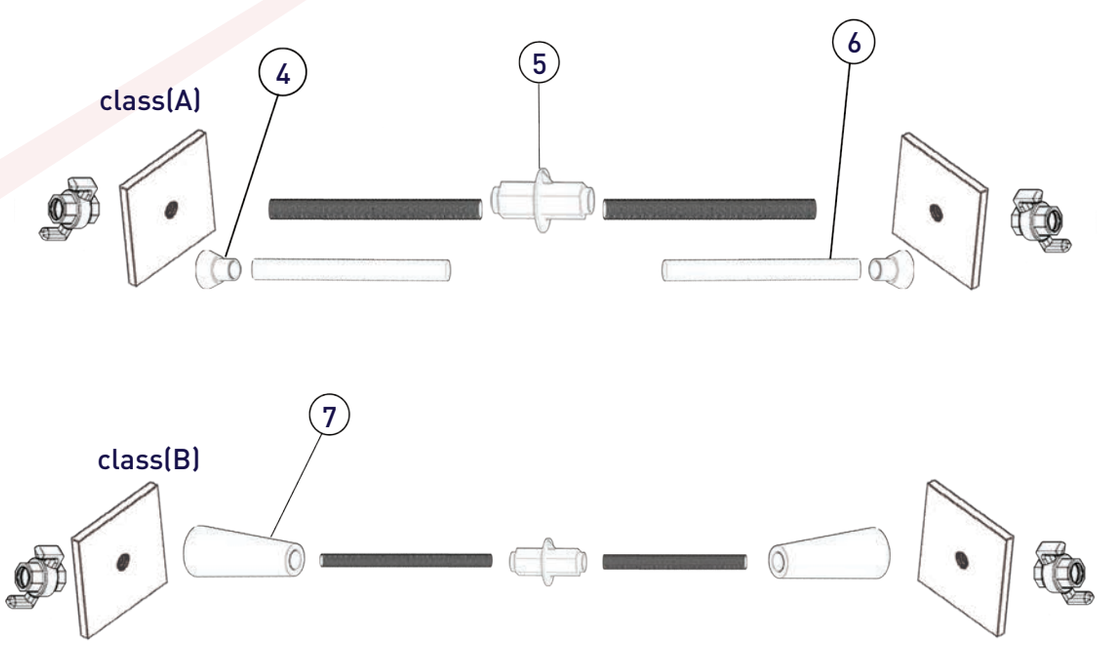
DW 16-Tie System

The Acrow DW 16-Tie system is primarily used with the panels requiring minimum number of ties. Moreover, the basis for the assembly is the continuous thread high tensile steel rod. Finally, the wing nut is designed to freely spin on the fast threaded rod in order to effectively speed the time for the formwork erection and stripping.

Thru DW 16 Tie System: Re - Usable



Water seal DW 16 Tie System

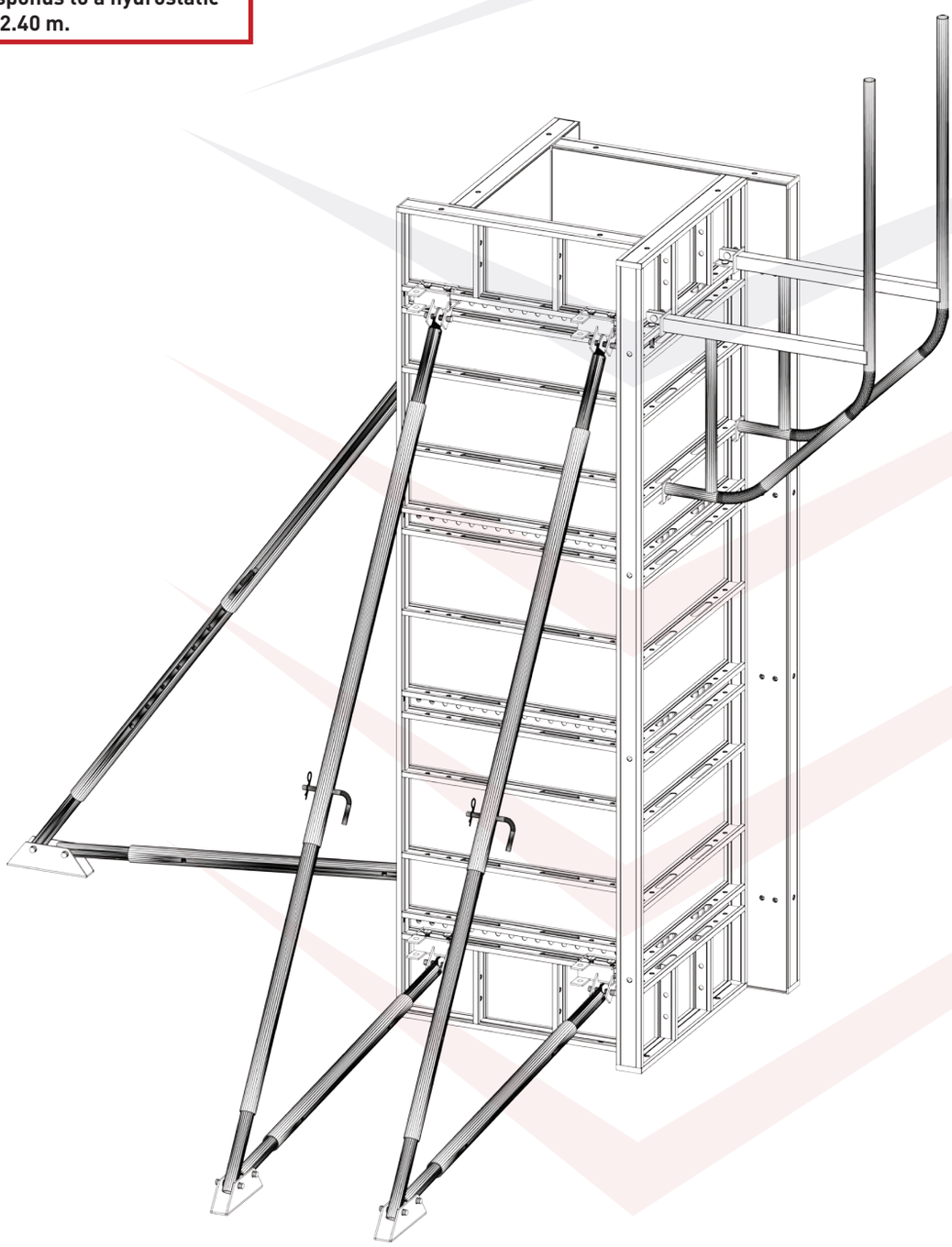


1- D W16 tie rod 2- Steel washer 3- Wing nut 4- Plastic cone 5-Water barrier 6- Palstic tube 7- Steel cone

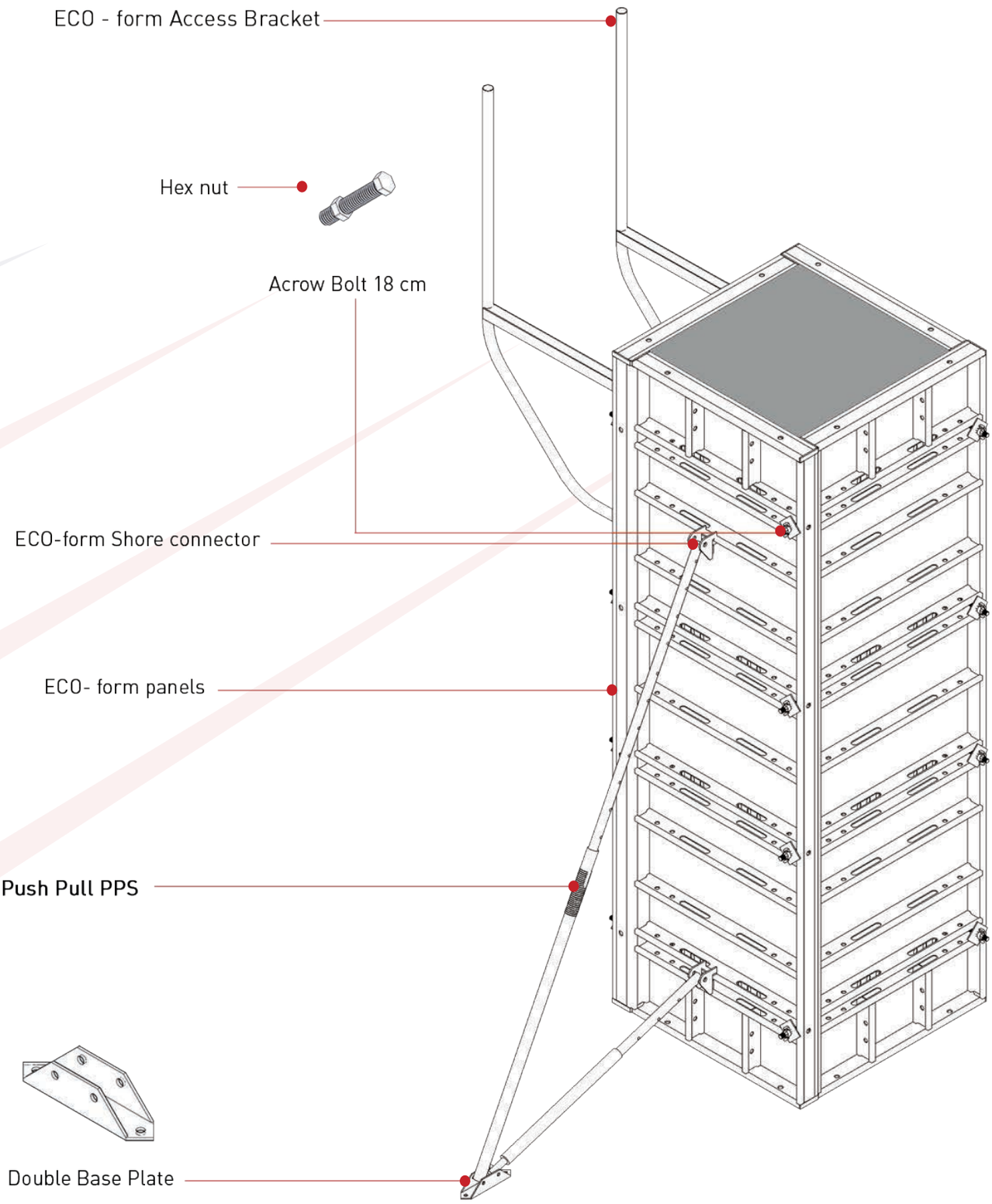
Column Formwork

Acrow Eco-form is simply a light weight panel formwork that is used for the walls and columns formwork with different dimensions.

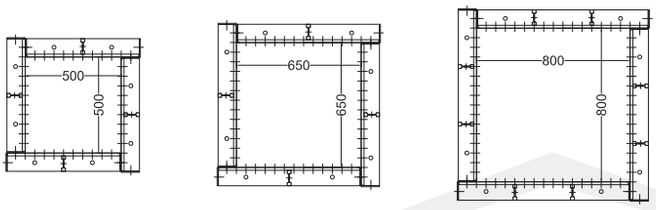
Permitted fresh-concrete pressure is 60 kN/m², where the concrete density is 25kN/m², where it corresponds to a hydrostatic pour height of 2.40 m.

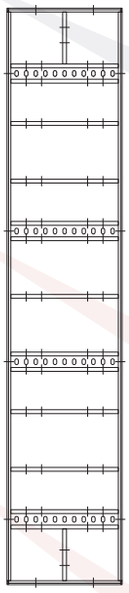
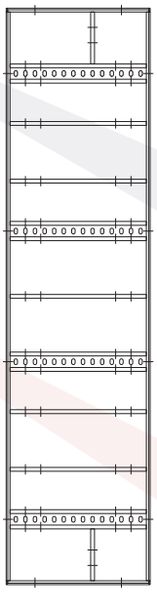
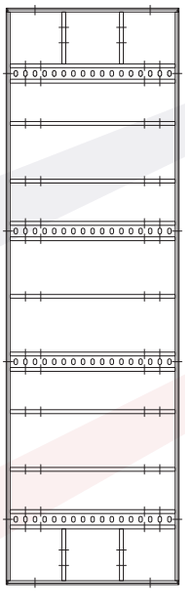
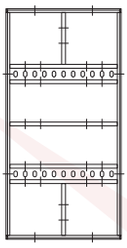
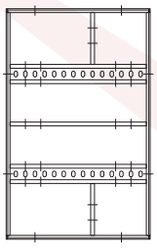
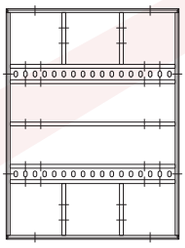
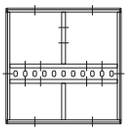
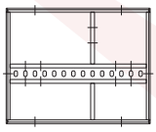
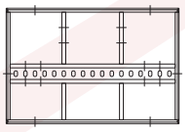


Column Overview



Column Panels Overview



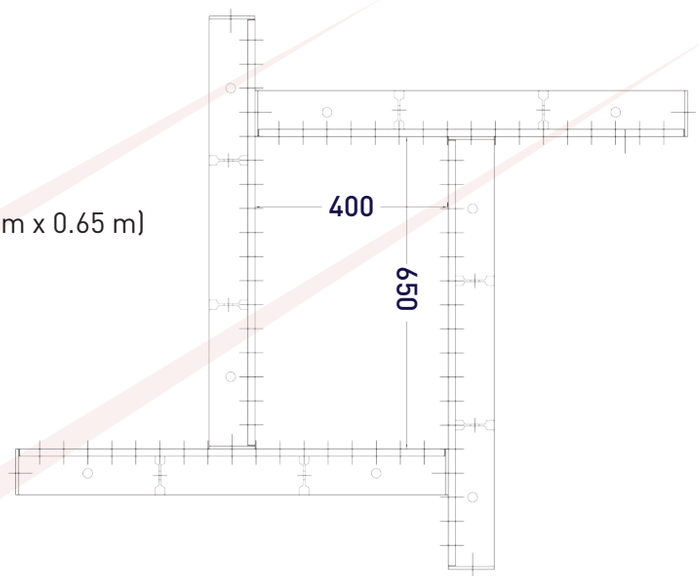
HEIGHT [cm]	WIDTH [cm] 60	75	90
300			
120			
60			

With ECO-Form Column Panels

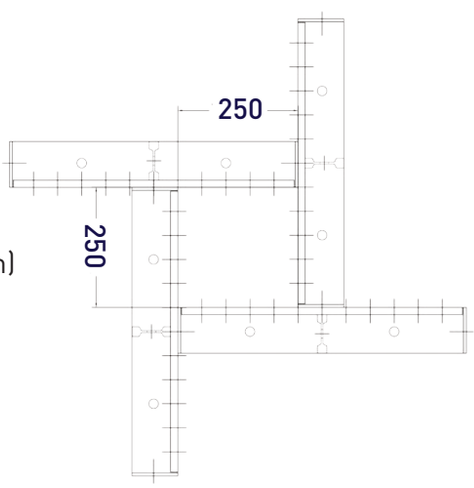
The 5.0 cm hole-grid enables forming different sizes of columns. It has a cross section up to 80 cm * 80 cm. Moreover, combining panels with heights 3.00 m, 1.20 m and 0.60 m could enable achieving different height grids.

Eco-Form panel (EFCP) 3.00m x 0.90m

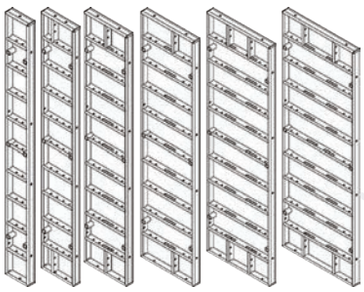
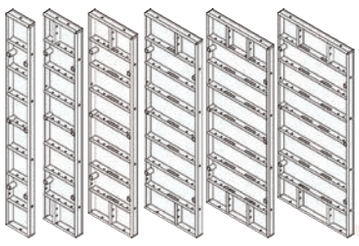
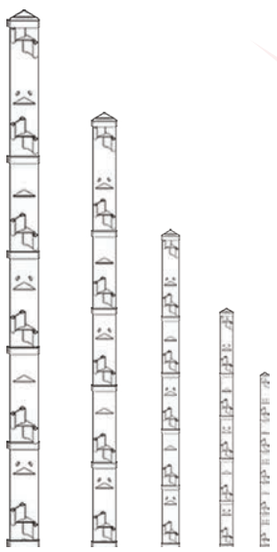
Example: Column (0.4 m x 0.65 m)



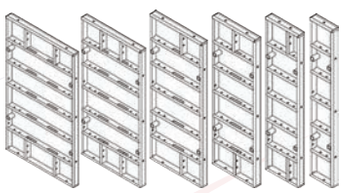
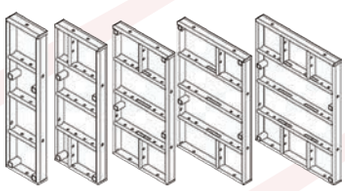
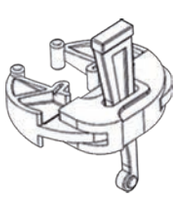
Example: Column (0.25 m x 0.25m)



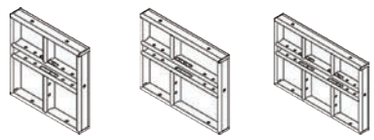
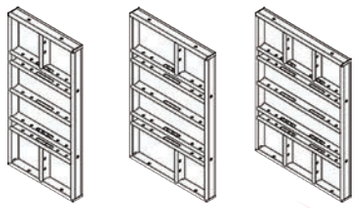
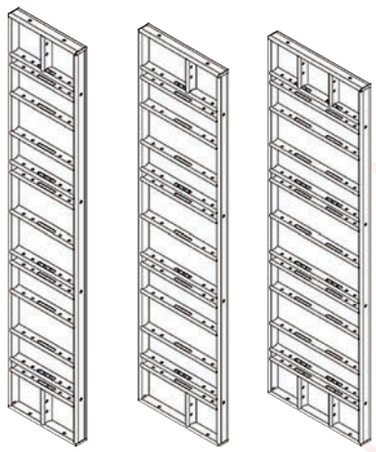
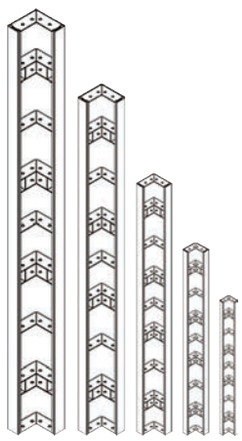
Components

Item	Description	Code no.	Weight (kg)
	ECO-form panel 3.00 × 0.30 m 3.00 × 0.45 m 3.00 × 0.60 m 3.00 × 0.75 m 3.00 × 0.90 m 3.00 × 1.05 m	6221177057633 6221177057640 6221177057657 6221177057664 6221177054892 6221177057671	45.30 56.70 65.54 75.46 86.00 95.80
	ECO-form panel 2.40 × 0.30 m 2.40 × 0.45 m 2.40 × 0.60 m 2.40 × 0.75 m 2.40 × 0.90 m 2.40 × 1.05 m	6221177057510 6221177057527 6221177057534 6221177057541 6221177057558 6221177057565	28.00 35.92 41.62 47.86 55.30 61.55
	ECO-form external corner angle (EFXC) 0.60 m (EFXC) 1.80 m (EFXC) 1.20 m (EFXC) 2.40 m (EFXC) 3.0 m	6221177057763 6221177057770 6221177057961 6221177057787 6221177057794	3.90 12.60 7.80 16.00 19.70

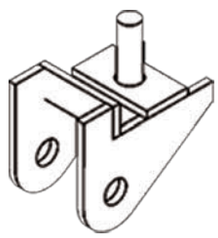
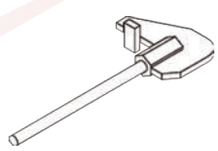
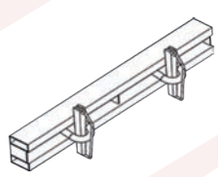
Components

Item	Description	Code no.	Weight (kg)
	ECO-form panel 1.80 × 0.30 m 1.80 × 0.45 m 1.80 × 0.60 m 1.80 × 0.75 m 1.80 × 0.90 m 1.80 × 1.05 m	 6221177057510 6221177057527 6221177057534 6221177057541 6221177057558 6221177057565	 28.00 35.92 41.62 47.86 55.30 61.55
	ECO-form panel 1.20 × 0.30 m 1.20 × 0.45 m 1.20 × 0.60 m 1.20 × 0.75 m 1.20 × 0.90 m	 6221177057466 6221177057473 6221177057480 6221177057497 6221177057503	 19.60 25.70 29.70 34.20 40.00
	ECO-form Coupler (EFC)	6221177054908	2.70

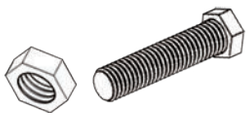
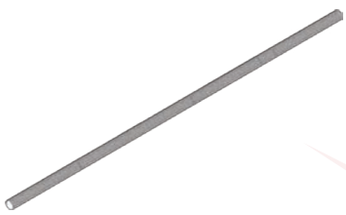
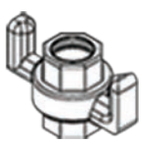
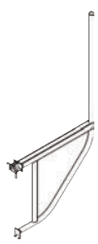
Components

Item	Description	Code no.	Weight (kg)
	ECO-form column panel 0.60X0.60 m 0.60X0.75 m 0.60X0.90 m	 6221177057688 6221177057695 6221177057701	 19.20 22.50 27.00
	ECO-form column panel 1.20X0.60 m 1.20X0.75 m 1.20X0.90 m	 6221177055233 6221177057718 6221177057725	 33.50 39.50 46.50
	ECO-form column panel 3.00X0.60 m 3.00X0.75 m 3.00X0.90 m	 6221177057732 6221177057749 6221177057756	 74.00 86.50 99.00
	ECO-form internal corner panel (EFIC) 0.60x0.30x0.30 m (EFIC) 1.20x0.30x0.30 m (EFIC) 1.80x0.30x0.30 m (EFIC) 2.40x0.30x0.30 m (EFIC) 3.00x0.30x0.30 m	 6221177057800 6221177057954 6221177057817 6221177057824 6221177057831	 18.63 33.80 50.80 61.60 73.00

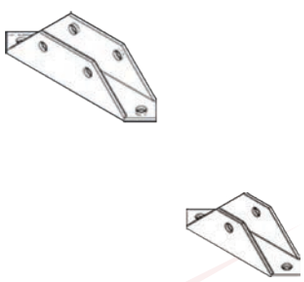
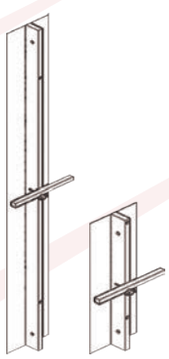
Components

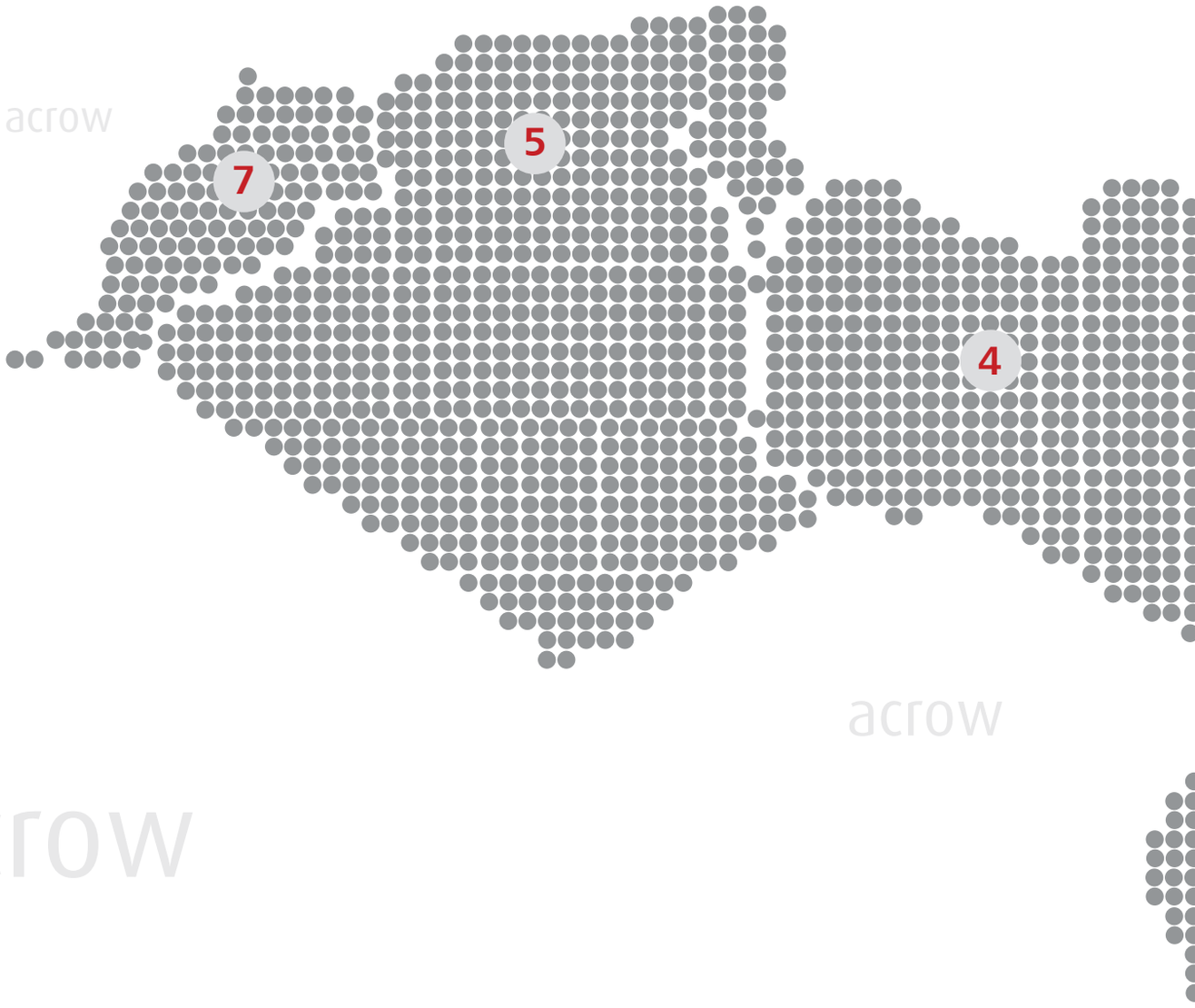
Item	Description	Code no.	Weight (kg)
	ECO-form Shore connector	6221177054922	2.687
	ECO-form Stop End Tie	6221177057862	2.70
	Compensation Waler	6221177057848	11.30
	Rivet Pin = 17 mm	6221177012298	0.25
	Spring Clip large Spring Clip small	6221177022617 6221177022624	0.03 0.01

Components

Item	Description	Code no.	Weight (kg)
	Acrow bolt L= 15 cm dvd	6221177000035	0.30
	Tie rod DVD L=0.75 mm L=1.00 mm L=1.25 mm L=1.50 mm L=1.75 mm L=2.00 mm L=2.50 mm L=3.00 mm L=3.50 mm L=4.00 mm L=5.00 mm L=6.00 mm	 6221177023270 6221177023294 6221177023300 6221177023317 6221177023324 6221177023331 6221177023355 6221177023362 6221177023379 6221177023386 6221177023409 6221177023423	 1.19 1.58 1.98 2.37 2.77 3.16 3.95 4.74 5.50 6.32 7.90 9.48
	Wing Nut dvd	6221177011383	0.35
	Steel Washer	6221177027513	1.10
	ECO-Form Access Bracket	6221177054915	10.80

Components

Item	Description	Code no.	Weight (kg)
	Double Base Plate Single Base Plate	6221177000813 6221177000882	2.40 1.45
	(TFFP) 3.30 X 0.40 M (TFFP) 1.20 X 0.40 M	6221177034153 6221177034160	58.00 24.00
	Stacking Device	6221177054649	7.00
	Push Pull PPS 131 Push Pull PPS 132 Push Pull PPS 161 Push Pull PPS 162 Push Pull PPS 201 Push Pull PPS 202 Push Pull PPS 252 Push Pull PPS 302 Push Pull PPS 352	6221177014254 6221177014261 6221177014278 6221177014285 6221177014292 6221177014308 6221177014315 6221177014322 6221177014339	8.60 11.40 9.80 12.40 13.10 13.30 15.00 19.00 23.10



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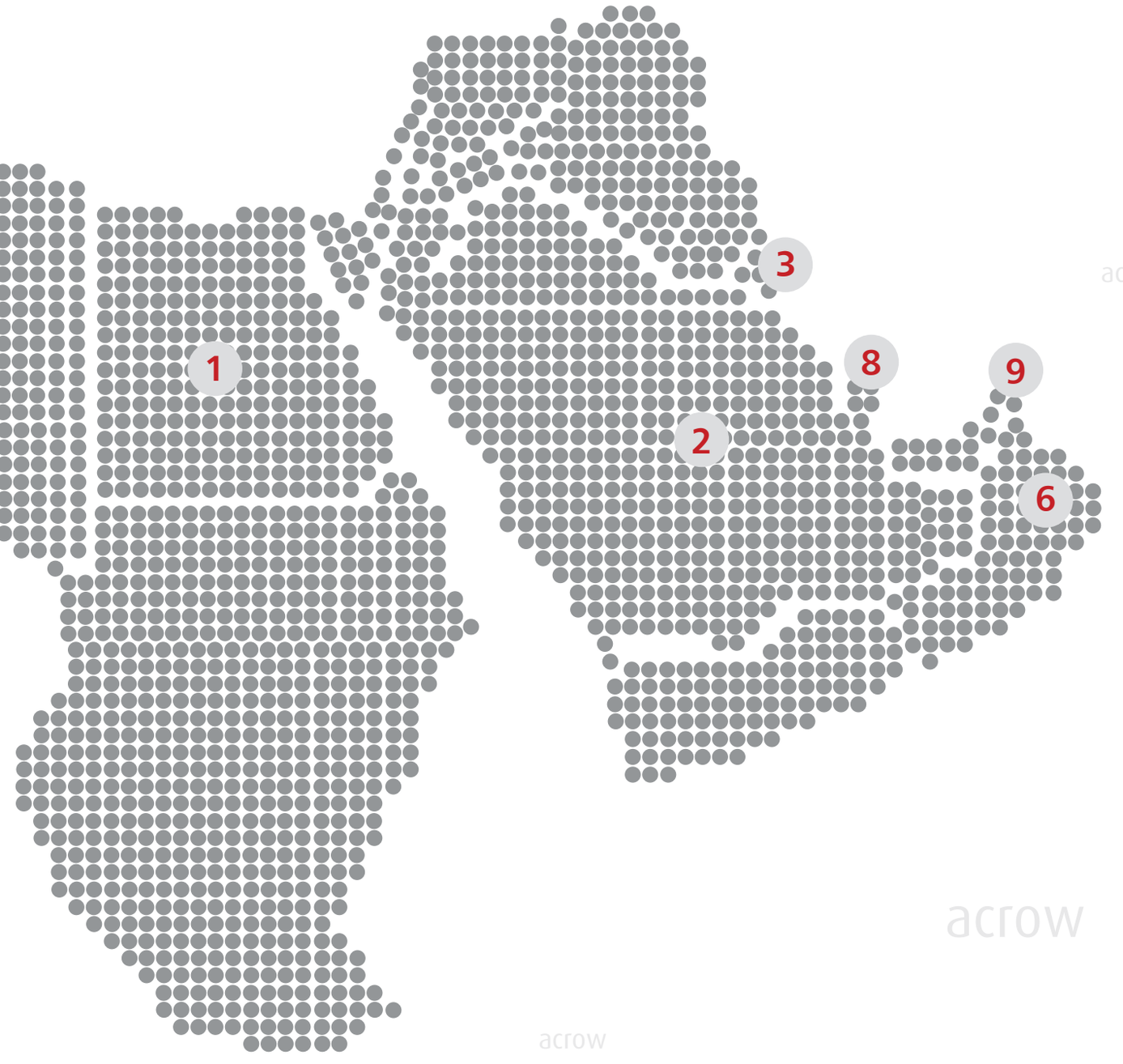
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